Case 4: Test of significance of the difference between two sample means.

$$\text{Test Statistic } z = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Note:

1. If the samples are drawn from the same population, that is, $\sigma = \sigma_1 = \sigma_2$, then
$$z = \frac{\overline{x_1} - \overline{x_2}}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$
2. If σ_1 and σ_2 are not known, then σ_1 and σ_2 can be approximated by the sample S.D.'s s_1 and s_2. Now,
$$z = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

The means of two large samples of 1000 and 2000 members are 67.5 inches and 68 inches respectively. Can the samples be regarded as drawn from the same population of S.D. 2.5 inches?

Soln:-

Null Hypothesis (H_0):- The samples have been drawn from the same population of standard deviation 2.5.
$(\mu_1 = \mu_2)$

Alternative Hypothesis (H_1): The samples are not drawn from the population of standard deviation 2.5

The hypothesis represents a two sided test

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Here $\bar{x}_1 = 67.5$; $\bar{x}_2 = 68$; $\sigma = 2.5$; $n_1 = 1000$
$n_2 = 2000$

$$z = \frac{67.5 - 68}{2.5\sqrt{\frac{1}{1000} + \frac{1}{2000}}} = -5.1$$

From the tabluar column

$$z_\alpha = 1.96$$

$\therefore |z| > |z_\alpha| \Rightarrow$ Reject Null Hypothesis.

In a survey of buying habits, 400 women shoppers are chosen at random in super market 'A' located in a certain section of the city. Their average weekly food expenditure is Rs. 250 with a standard deviation of Rs. 40. For 400 women shoppers chosen at random in super market 'B' in another section of the city, the average weekly food expenditure is Rs. 220 with a standard deviation of Rs. 55. Test at 1% level of significance whether the average weekly food expenditure of the two populations of shoppers are equal.

Soln:- Null Hypothesis (H_0) :- The average weekly food expenditure of the two population of shoppers are equal ($\mu_1 = \mu_2$)

Alternative Hypothesis (H_1): The average weekly food expenditure of the two population differs significantly. ($\mu_1 \neq \mu_2$)

This hypothesis represents a two tailed test.

Test statistic
$$z = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$
(Here the population SD is not available)

Here $\overline{x_1} = 250$; $\overline{x_2} = 220$; $n_1 = 400$; $n_2 = 400$; $s_1 = 40$; $s_2 = 55$

Thus
$$z = \frac{250 - 220}{\sqrt{\frac{40^2}{400} + \frac{55^2}{400}}} = \frac{30}{\sqrt{4 + 7.5625}} = 3.1373$$

From the tabular column $Z_\alpha = 2.58$

Since $|Z| > Z_\alpha$ we reject the null hypothesis.

The average hourly wage of a sample of 150 workers in a plant 'A' was Rs. 2·56 with a standard deviation of Rs. 1·08. The average wage of a sample of 200 workers in plant 'B' was Rs. 2·87 with a standard deviation of Rs. 1·28. Can an applicant safely assume that the hourly wages paid by plant 'B' are higher than those paid by plant 'A'?

Soln:- Null Hypothesis (H_0): There is no significant difference between between the mean level of wages ($\mu_1 = \mu_2$) of workers in Plant A and Plant B.

Alternative Hypothesis (H_1): Plant B is paying higher than ($\mu_2 > \mu_1$) Plant A.

This Hypothesis represents a ~~two~~ left tailed test.

The test Statistic $Z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$ (Here population variance not available)

$\bar{x}_1 = 2.56$; $\bar{x}_2 = 2.87$; $n_1 = 150$; $n_2 = 200$; $s_1 = 1.08$; $s_2 = 1.28$

$$Z = \frac{2.56 - 2.87}{\sqrt{\dfrac{(1.08)^2}{150} + \dfrac{(1.28)^2}{200}}} = -2.46$$

From the tabular column $Z_{0.05} = -1.645$

Since $|Z| > |Z_{0.05}|$, we reject the null hypothesis at 5% level of significance.